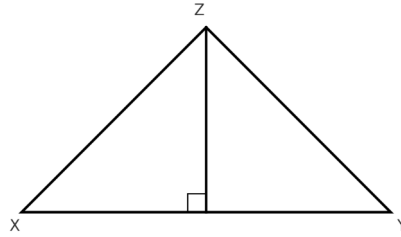


Math Hard Question 2

1

★ Mark for Review

ABC



In the figure shown, the measure of angle Y is 38° . The length of XY is 12 meters, and the length of YZ is 5 meters. What is the area, in square meters, of triangle XYZ ?

A. 30

B. 60

C. $30 \sin 38^\circ$ D. $60 \sin 38^\circ$

2

★ Mark for Review

ABC

A research manager selected two random samples to estimate the average lifespan of a certain type of light bulb (in hours). Based on the first sample, the research manager estimated that the average lifespan of this type of light bulb is 800 hours, with a margin of error of 20 hours.

Based on the second sample, the research manager estimated that the average lifespan of this type of light bulb is 810 hours, with a margin of error of 35 hours.

Assuming the margins of error were calculated in the same way, which of the following best explains why the first sample obtained a smaller margin of error than the second sample?

- A. The first sample contained fewer light bulbs than the second sample.
- B. The first sample contained more light bulbs than the second sample.
- C. The first sample had a shorter average lifespan than the second sample.
- D. The first sample had a longer average lifespan than the second sample.

3

★ Mark for Review

ABC

The function $p(x)$ is defined by:

$$p(x) = a((x - 3)^2 + b)((x - 3)^2 - c),$$

where a , b , and c are constants. The graph of $y = p(x)$ passes through the points $(1, 25)$ and $(-1, 52)$.

What is the value of $p(5) + p(7)$?

- A. 25
- B. 27
- C. 52
- D. 77

4

★ Mark for Review

ABC

In the given function, b is a positive constant. For every increase in the value of x by 1, the value of $f(x)$ increases by $c\%$, where $0 < c < 100$.

Which expression gives the value of c in terms of b ?

$$f(x) = 77(b)^x$$

A. $1 + \frac{b}{100}$

B. $b + 100$

C. $100(b + 1)$

D. $100(b - 1)$

5

★ Mark for Review

ABC

On average, a certain tree grows 87 centimeters every m months. At this rate, which expression represents the number of centimeters, on average, the tree grows every k years?

A. $\frac{29m}{4k}$

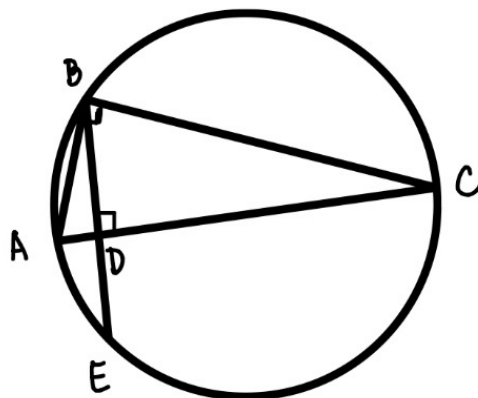
B. $\frac{29k}{4m}$

C. $\frac{1044m}{k}$

D. $\frac{1044k}{m}$

6

★ Mark for Review

ABC

In the figure shown, points A , B , C , and E lie on the circle, and $AB < BC$. Segment AC is perpendicular to segment BE at point D , and $BD = \sqrt{346}$. The diameter of the circle is 175. If $\frac{CD}{AD} = r$, what is the value of r ?

7

★ Mark for Review

ABC

In the given equation, a and k are constants, where $k > 5a$. The sum of the solutions to the equation is $3k + 32$. What is the value of a ?

$$(x - k)^2 = (k - 5a)(x - k)$$

A. $-\frac{32}{5}$

B. $\frac{32}{5}$

C. $-\frac{5}{32}$

D. $\frac{5}{32}$

8

★ Mark for Review

ABC

The equation $N(m) = 63(Q)^{m/6}$ gives the predicted population $N(m)$, in thousands, of a certain bacteria colony m minutes after the initial measurement, where Q is a constant greater than 1. The predicted population increases by $p\%$ every 180 seconds.

What is the value of p in terms of Q ?

A. $100(Q^{30} - 1)$

B. $100(Q^{\frac{1}{2}} - 1)$

C. $100(Q^{30} + 1)$

D. $100(Q^{1.5} + 1)$

9

★ Mark for Review

ABC

In the given equation, m and n are constants. The sum and product of the solutions of the equation are 8 and $\frac{7}{2}$, respectively.

What is the value of $\frac{m}{n}$?

$$(x - m)^2 = (m + 2n)(2x - n)$$

A. $-\frac{1}{3}$

B. $\frac{1}{3}$

C. -3

D. 3